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QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 1.0.0.11
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 1.0.0.1

C:\Documents and Settings\User>
```

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QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 129.23.0.2
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 129.23.0.1

C:\Documents and Settings\User>
```

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 29.0.0.11
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 29.0.0.1

C:\Documents and Settings\User>
```

```
QEMU (Win-4)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 212.25.0.11
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 212.25.0.1

C:\Documents and Settings\User>
```

```

R1
!
ip dhcp excluded-address 1.0.0.1 1.0.0.10
ip dhcp excluded-address 29.0.0.1 29.0.0.10
ip dhcp excluded-address 212.25.0.1 212.25.0.10
ip dhcp excluded-address 129.23.0.1 129.23.0.10
!
ip dhcp pool SALE
network 1.0.0.0 255.0.0.0
default-router 1.0.0.1
!
ip dhcp pool HR

```

```

R2
R2#sh ip int brief
Interface      IP-Address      OK? Method Status  Prot
Ethernet0/0    172.50.0.2      YES TFTP  up      up
Ethernet0/1    12.0.0.1        YES TFTP  up      up
Ethernet0/2    222.215.0.1     YES TFTP  up      up
Ethernet0/3    62.0.0.1        YES TFTP  up      up

```

```

R3
*Aug  3 15:15:38.043: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int brief
Interface      IP-Address      OK? Method Status  Prot
Ethernet0/0    12.0.0.2        YES TFTP  up      up
Ethernet0/1    29.0.0.1        YES TFTP  up      up
Ethernet0/2    unassigned      YES TFTP  down    down
Ethernet0/3    unassigned      YES TFTP  down    down
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#

```

```

R4
R4(config)#exit
R4#sh
*Aug  3 15:15:49.802: %SYS-5-CONFIG_I: Configured from console by console
R4#sh ip int brief
Interface      IP-Address      OK? Method Status  Prot
Ethernet0/0    222.215.0.2     YES TFTP  up      up
Ethernet0/1    129.23.0.1      YES TFTP  up      up
Ethernet0/2    unassigned      YES TFTP  down    down
Ethernet0/3    unassigned      YES TFTP  down    down

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R5
Interface      IP-Address      OK? Method Status  Prot
Ethernet0/0    62.0.0.2        YES TFTP  up      up
Ethernet0/1    212.25.0.1      YES TFTP  up      up
Ethernet0/2    unassigned      YES TFTP  down    down
Ethernet0/3    unassigned      YES TFTP  down    down

```

```

R1
ip dhcp excluded-address 1.0.0.1 1.0.0.10
ip dhcp excluded-address 29.0.0.1 29.0.0.10
ip dhcp excluded-address 212.25.0.1 212.25.0.10
ip dhcp excluded-address 129.23.0.1 129.23.0.10
!
ip dhcp pool SALE
network 1.0.0.0 255.0.0.0
default-router 1.0.0.1
!
ip dhcp pool HR
network 29.0.0.0 255.0.0.0
default-router 29.0.0.1
!
ip dhcp pool PRO
network 212.25.0.0 255.255.255.0
default-router 212.25.0.1
!
ip dhcp pool MGMT
network 129.23.0.0 255.255.0.0
default-router 129.23.0.1

```

```
R1
Gateway of last resort is not set

1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 1.0.0.0/8 is directly connected, Ethernet0/1
L 1.0.0.1/32 is directly connected, Ethernet0/1
S 12.0.0.0/8 [1/0] via 172.50.0.2
S 29.0.0.0/8 [1/0] via 172.50.0.2
S 62.0.0.0/8 [1/0] via 172.50.0.2
S 129.23.0.0/16 [1/0] via 172.50.0.2
S 172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.50.0.0/16 is directly connected, Ethernet0/0
L 172.50.0.1/32 is directly connected, Ethernet0/0
S 212.25.0.0/24 [1/0] via 172.50.0.2
S 222.215.0.0/24 [1/0] via 172.50.0.2
R1#

R2
+ - replicated route, % - next hop override, p - override
Gateway of last resort is not set

S 1.0.0.0/8 [1/0] via 172.50.0.1
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.0.0.0/8 is directly connected, Ethernet0/1
L 12.0.0.1/32 is directly connected, Ethernet0/1
S 29.0.0.0/8 [1/0] via 12.0.0.2
S 62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 62.0.0.0/8 is directly connected, Ethernet0/3
L 62.0.0.1/32 is directly connected, Ethernet0/3
S 129.23.0.0/16 [1/0] via 222.215.0.2
S 172.50.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.50.0.0/16 is directly connected, Ethernet0/0
L 172.50.0.2/32 is directly connected, Ethernet0/0
S 212.25.0.0/24 [1/0] via 62.0.0.2
S 222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
C 222.215.0.0/24 is directly connected, Ethernet0/2
L 222.215.0.1/32 is directly connected, Ethernet0/2
R2#
R2#
R2#

R3
S 1.0.0.0/8 [1/0] via 12.0.0.1
C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.0.0.0/8 is directly connected, Ethernet0/0
L 12.0.0.2/32 is directly connected, Ethernet0/0
S 29.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 29.0.0.0/8 is directly connected, Ethernet0/1
L 29.0.0.1/32 is directly connected, Ethernet0/1
S 62.0.0.0/8 [1/0] via 12.0.0.1
S 129.23.0.0/16 [1/0] via 12.0.0.1
S 172.50.0.0/16 [1/0] via 12.0.0.1
S 212.25.0.0/24 [1/0] via 12.0.0.1
S 222.215.0.0/24 [1/0] via 12.0.0.1
R3#
R3#

R4
S 1.0.0.0/8 [1/0] via 222.215.0.1
S 12.0.0.0/8 [1/0] via 222.215.0.1
S 29.0.0.0/8 [1/0] via 222.215.0.1
S 62.0.0.0/8 [1/0] via 222.215.0.1
S 129.23.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 129.23.0.0/16 is directly connected, Ethernet0/1
L 129.23.0.1/32 is directly connected, Ethernet0/1
S 172.50.0.0/16 [1/0] via 222.215.0.1
S 212.25.0.0/24 [1/0] via 222.215.0.1
S 222.215.0.0/24 is variably subnetted, 2 subnets, 2 masks
C 222.215.0.0/24 is directly connected, Ethernet0/0
L 222.215.0.2/32 is directly connected, Ethernet0/0
R4#
R4#
R4#
R4#
R4#
R4#
R4#
R4#
R4#
R4#

R5
S 1.0.0.0/8 [1/0] via 62.0.0.1
S 12.0.0.0/8 [1/0] via 62.0.0.1
S 29.0.0.0/8 [1/0] via 62.0.0.1
C 62.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 62.0.0.0/8 is directly connected, Ethernet0/0
L 62.0.0.2/32 is directly connected, Ethernet0/0
S 129.23.0.0/16 [1/0] via 62.0.0.1
S 172.50.0.0/16 [1/0] via 62.0.0.1
S 212.25.0.0/24 is variably subnetted, 2 subnets, 2 masks
C 212.25.0.0/24 is directly connected, Ethernet0/1
L 212.25.0.1/32 is directly connected, Ethernet0/1
S 222.215.0.0/24 [1/0] via 62.0.0.1
R5#
R5#
R5#
R5#
R5#
R5#
R5#
R5#
R5#
R5#
```